Technical Session

→ **Linear Relationship**: A straight line relationship between two variables.

Mathamatically, Straight Line Equation —

$$\boxed{Y = mX + C}$$

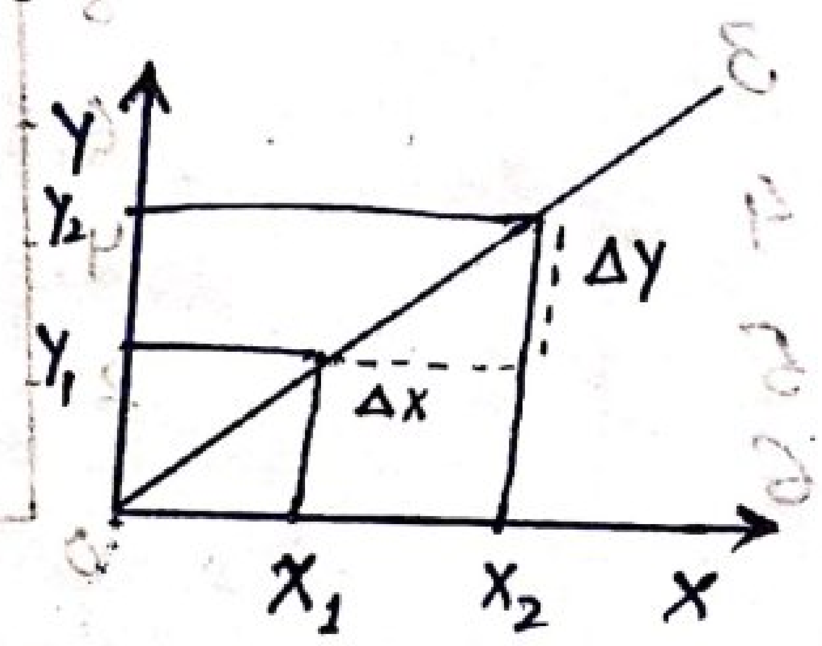
Positive Slope: When two variables are moving in the same direction.

Negative Slope: When two variables are moving in different direction

No Slope: Straight Line (No change)

Slope → $\frac{\Delta H}{\Delta D} = 0$ ⇒ Here, H is Hight, D is Direction.

Y is Height
X is Distance



$$Y_1 = mX_1 + C \text{ —— (i)}$$
$$Y_2 = mX_2 + C \text{ —— (ii)}$$
$$Y_2 - Y_1 = m(X_2 - X_1)$$
$$\Delta Y = m\Delta X$$

⇒ $\frac{\Delta Y}{\Delta X} = m$ (Slope) ⇒ Shows ΔY due to ΔX — 1
Steapness & direction — 2
+, −, No slope, undefined — 3

Slope

How much unit ΔY due to 1 additional unit ΔX.

3 Concept of Slope : —

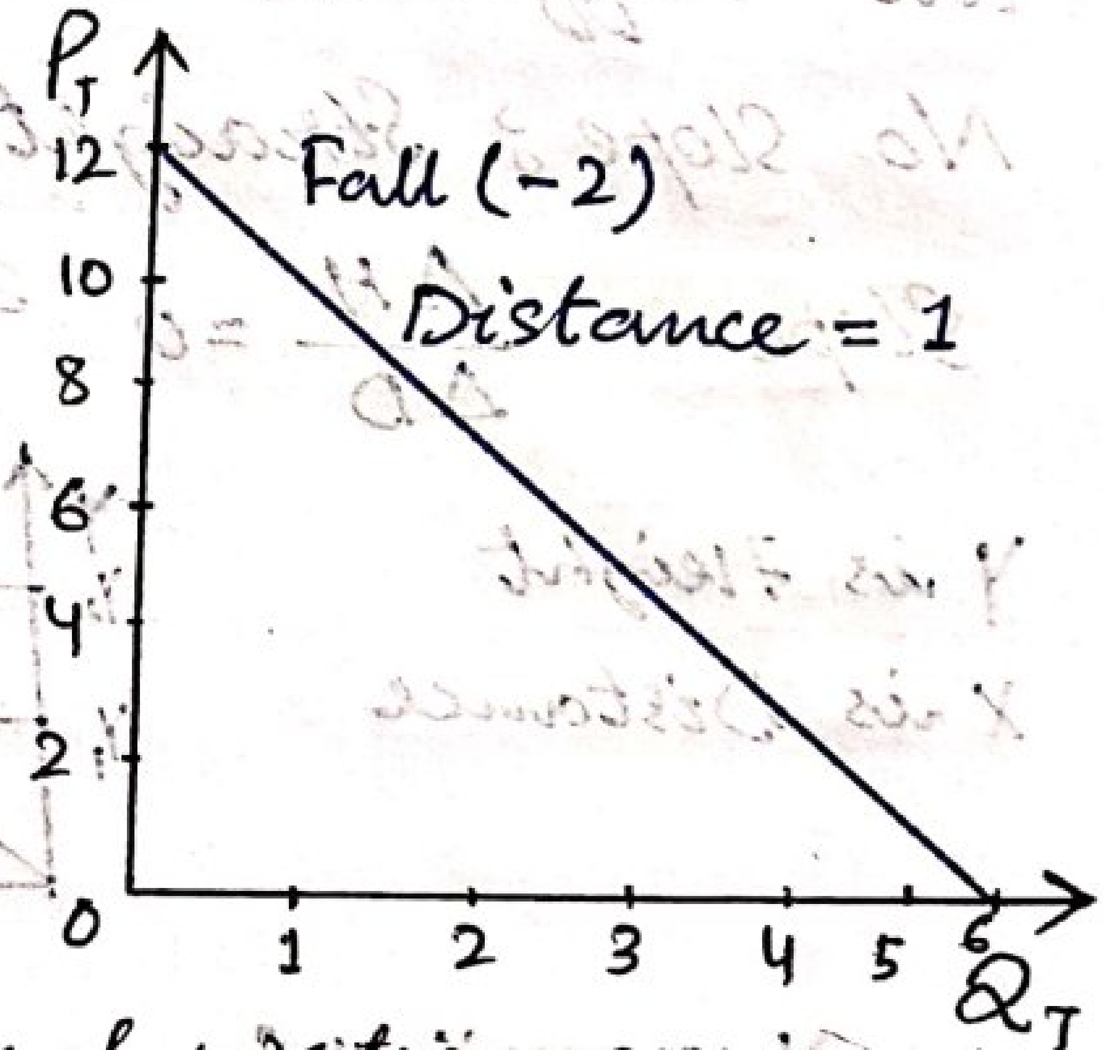
1) Unique Solution $\frac{10}{5} = x$
2) No solution / Undefined $\frac{10}{0} = x$
3) Infinity. $\frac{0}{0} = x$

Intercept

→ The value of a variable which not depend on a variable but on others. constants or etc.
e.g. Income is X and C is consumption, If $x=0$ then the consumption is Intercept, it may be $-$, $+$ or 0

Examples of Intercept →

Price (P_T)	Quantity (Q_T)
12	0
10	1
8	2
6	3
4	4
2	5
0	6



$$(\text{Slope})\ m = \frac{\Delta Y}{\Delta X} = \frac{\Delta P_T}{\Delta Q_T}$$

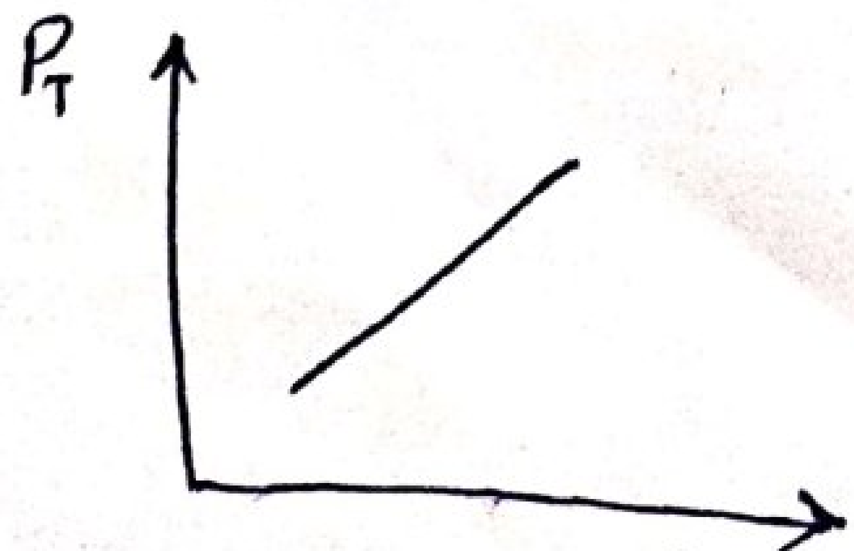
$$P_T = 12 - 2Q_T$$

[General ~~function~~ formulae of Demand function.
$P = a - bq$ ($b = m$ (slope))]

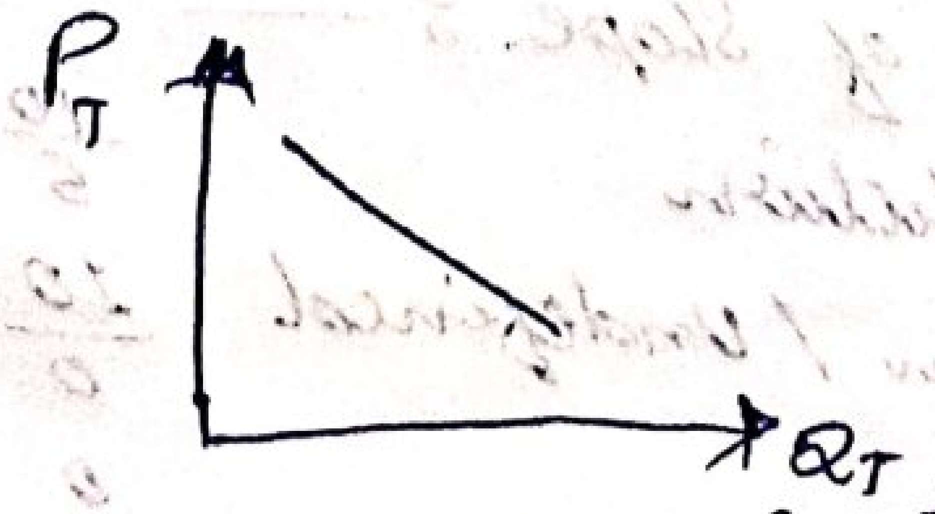
Interpretation

→ When prices reduces by 2 ruppees then the quantity (demand) increases by 1 additional unit.

- When slope is constant then the graph



Positive constant



Negative Constant

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→ Straight line Possibility ($Y = mX + c$)

	$c>0$	$c=0$	$c<0$
$m>0$	$m>0$, $c>0$	$m>0$, $c=0$	$m>0$, $c<0$
$m=0$	$m=0$, $c>0$	$m=0$, $c=0$	$m=0$, $c<0$
$m<0$	$m<0$, $c>0$	$m<0$, $c=0$	$m<0$, $c<0$

1. $m>0$, $c>0$

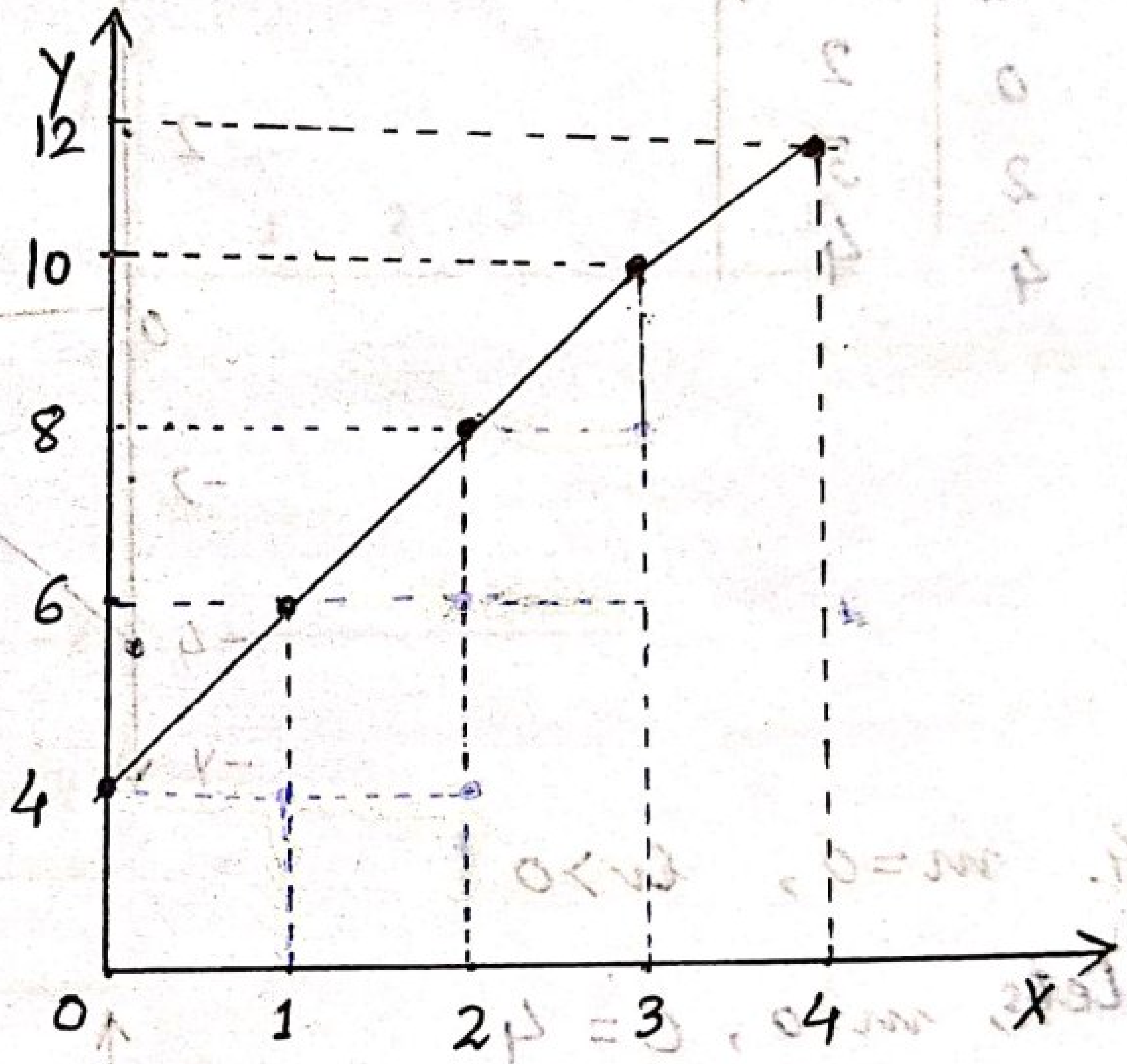
Lets, $m = +2$, $c = +4$

$Y = mX + c$

$= +2X + 4$

$Y = -4$

Y	X	m
+4	0	+2
6	1	+2
8	2	2
10	3	2
12	4	2



2. $m>0$, $c=0$

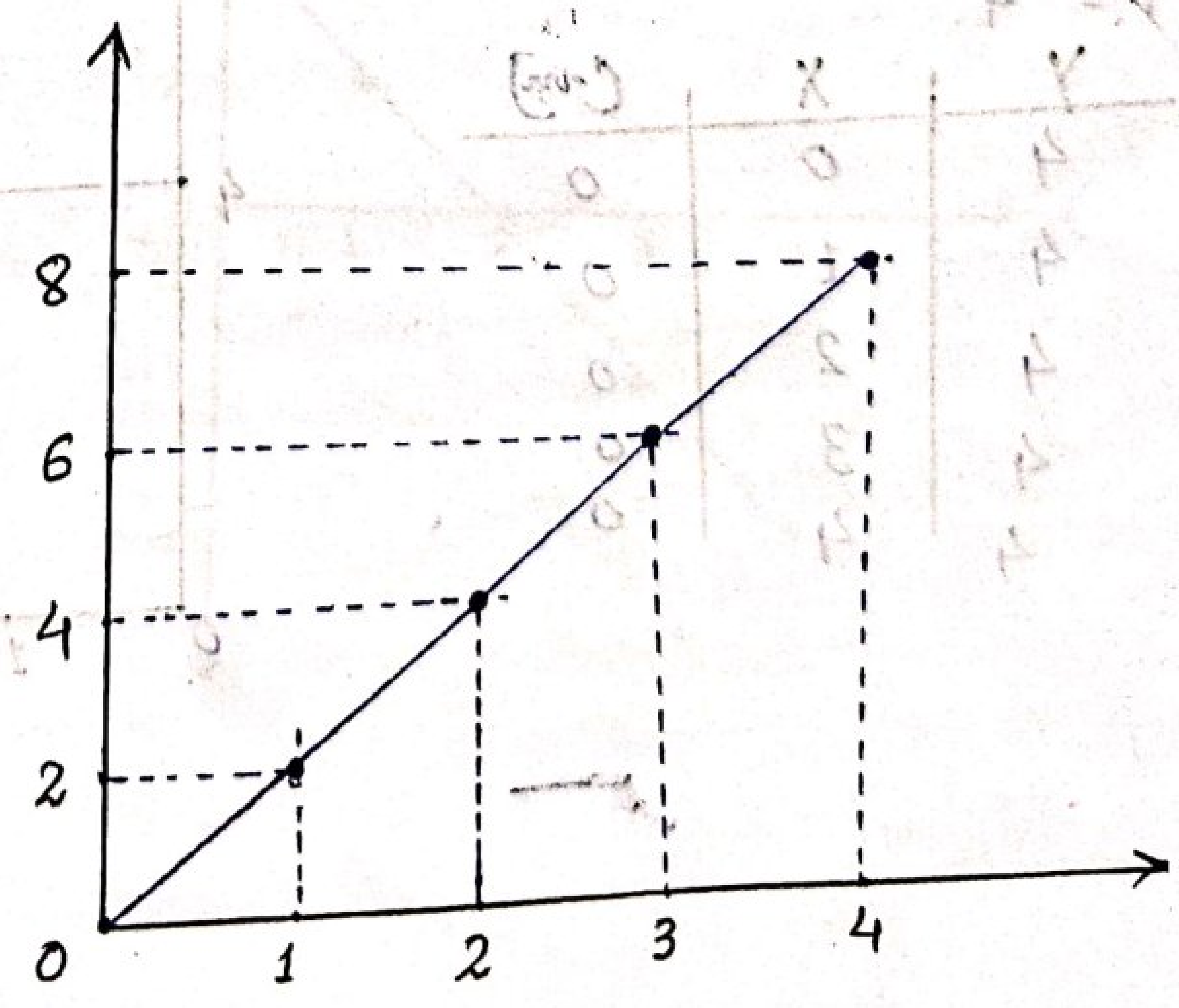
Lets, $m = 2$, $c = 0$

$Y = mX + c$

$= 2X + 0$

$Y = 0$

y	X	m
0	0	2
2	1	2
4	2	2
6	3	2
8	4	2



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Unit-I. Variable Parameter and Constant

Independent variable → X is Independent variable

Dependent variable → Y is the Dependent variable.

$[Y = mX + c] \rightarrow$ Linear

Changes the value of X consider, is the value of Y.

Intercept :- It is the value of one variable when the other variable are 0.

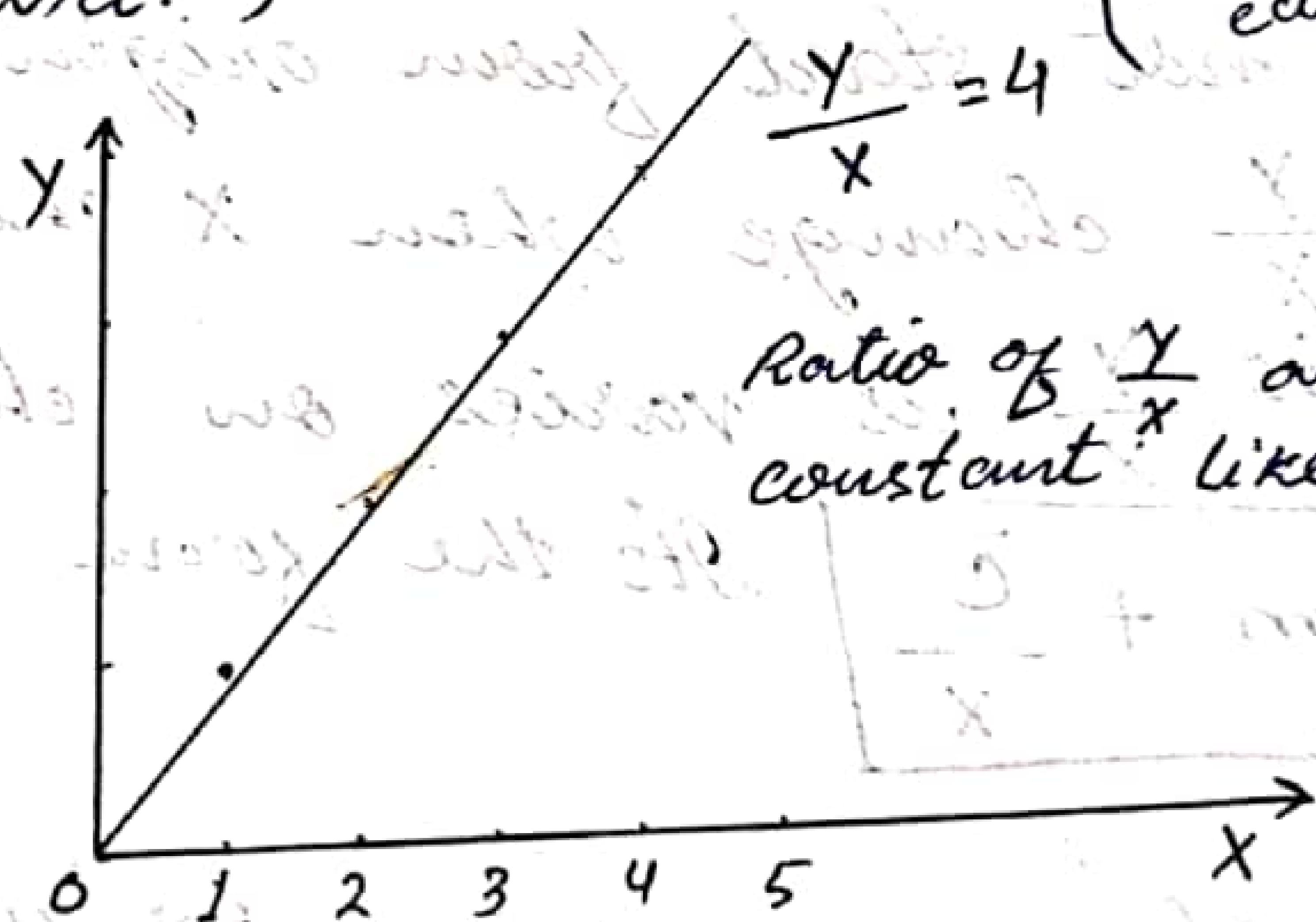
(will be used in consumption function) Proportion and Non Proportion

— Relationship —

Proportion relationship → It is relationship btn two variables where the ratio are always equal or constant. It also represent rate of ΔY with respect to ΔX. (will be used in development economics)

(wrt.)

X	Y
0	0
1	4
2	8
3	12
4	16
5	20



$\frac{Y}{X} = 4$

Ratio of $\frac{Y}{X}$ always be constant like 4.

Features of PR —

→ $\frac{Y}{X}$ or proportional relationship always be constant

→ $\frac{Y}{X}$, $\frac{\Delta Y}{\Delta X}$

→ Start from origin

→ Every line that divides Y axis and X axis equally is making 45° Angle (Regression model)

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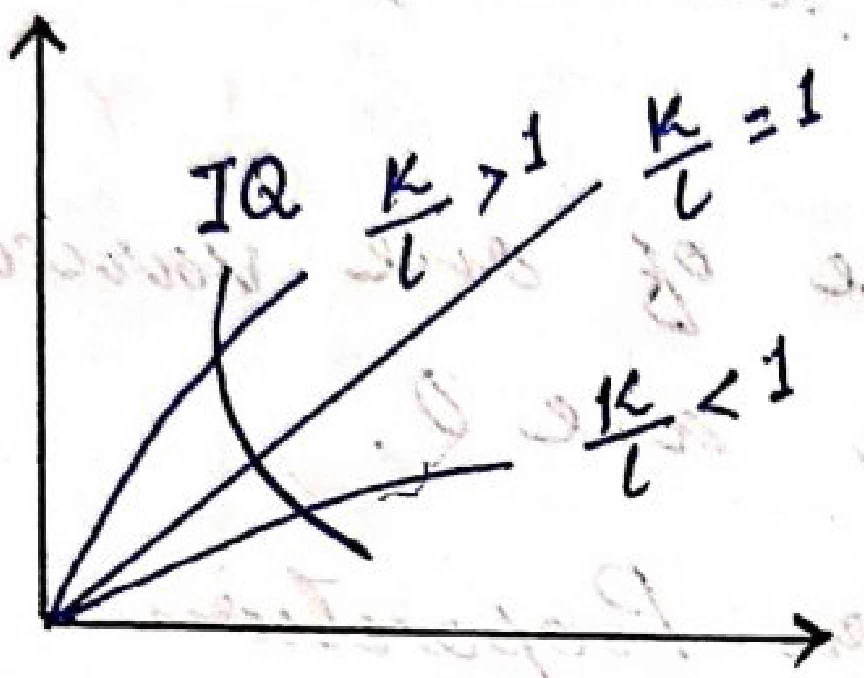
— Economy —

1. India – Labour abundant country ⇒ $\frac{K}{L}$ low
2. USA – Capital abundant " ⇒ $\frac{K}{L}$ High
3. Neutral Technology – $\frac{K}{L}$ used in the same proportion.



IQ – ISO – Quant.
Equal Output

(Linear) Non-proportion Relationship —

It means relationship between two variable not remain constant.

→ Features —

- $Y = mX + C$ [Where $C \gtrless 0$ we have non P.R.]
- It is not start from origin.
- Ratio $\frac{Y}{X}$ change when X change

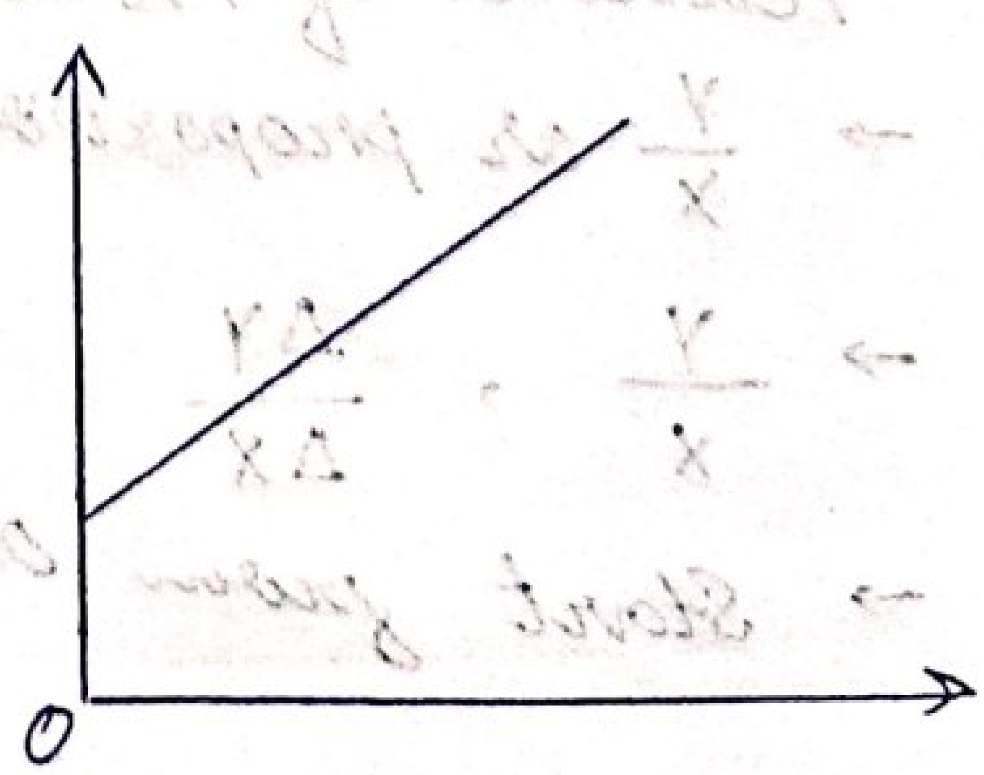
* In this $\frac{Y}{X}$ is varies on change in X value

$$\frac{Y}{X} = \boxed{m + \frac{C}{X}}$$ It's the form of NPR

$\frac{Y}{X}$ ratio shows average on per unit

e.g. $\frac{Y}{X} = \frac{60+70+80}{3} = 70$

average = per unit



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$\Delta \rightarrow$ Change $\rightarrow$ in economics $\rightarrow$ Marginal

$\frac{Y}{X} \rightarrow$ ratio $\rightarrow$ = $\rightarrow$ Average.

$\frac{C}{Q} \rightarrow$ If quantity or output locate in denominator then we never be pronounce it.

We say it Average cost.

$\frac{\Delta C}{\Delta Q} \rightarrow$ Marginal Cost...

$\rightarrow$ How much ~~output~~ change in cost due to 1 additional unit change in Quantity is called Marginal cost.

$\frac{\Delta Q}{\Delta L} \rightarrow$ How much output increases when labour increases by 1 additional unit. ~~of labour~~.

— Slope —

Various Possibilities

Case - 1. $m > 0$, $c > 0$

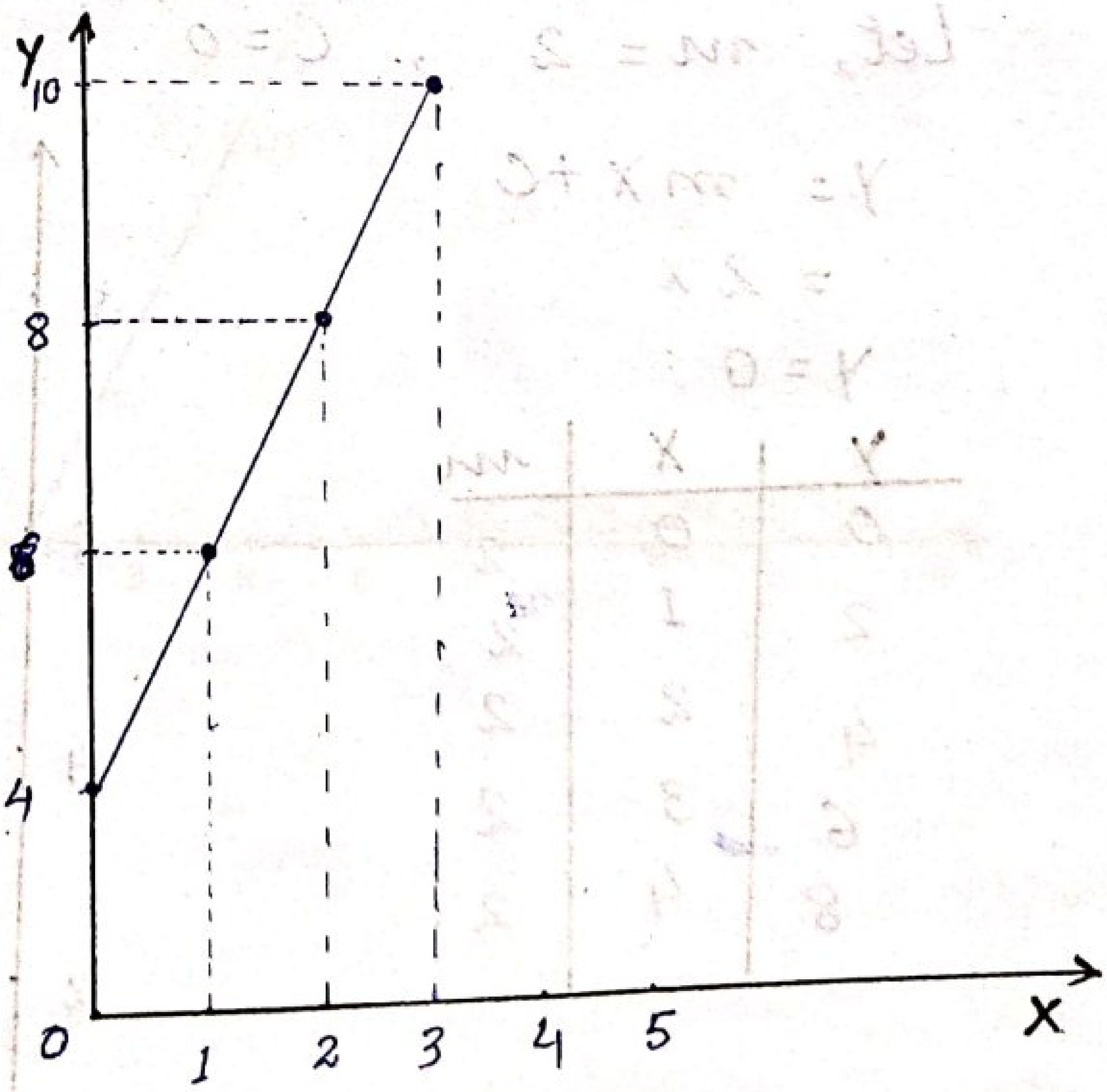
Lets, $m = 2$, $C = 4$

$Y = mX + C$

$= 2X + 4$

$Y = 4$

Y	X	m
4	0	2
6	1	2
8	2	2
10	3	2



Consumption and Income

$Y = mX + C$

$$\boxed{C = \bar{C} + cY}$$

Here, $\bar{C}$ = Autonomous Consumption, is Independent of Income.

$c = \frac{\Delta C}{\Delta Y}$ = Induced Consumption,

↳ Marginal Propensity to consume (MPC)

Income = Output

$$\boxed{0 < MPC < 1}$$

Propensity :- At which rate we buy something.

Case - 2. $m > 0$, $C = 0$

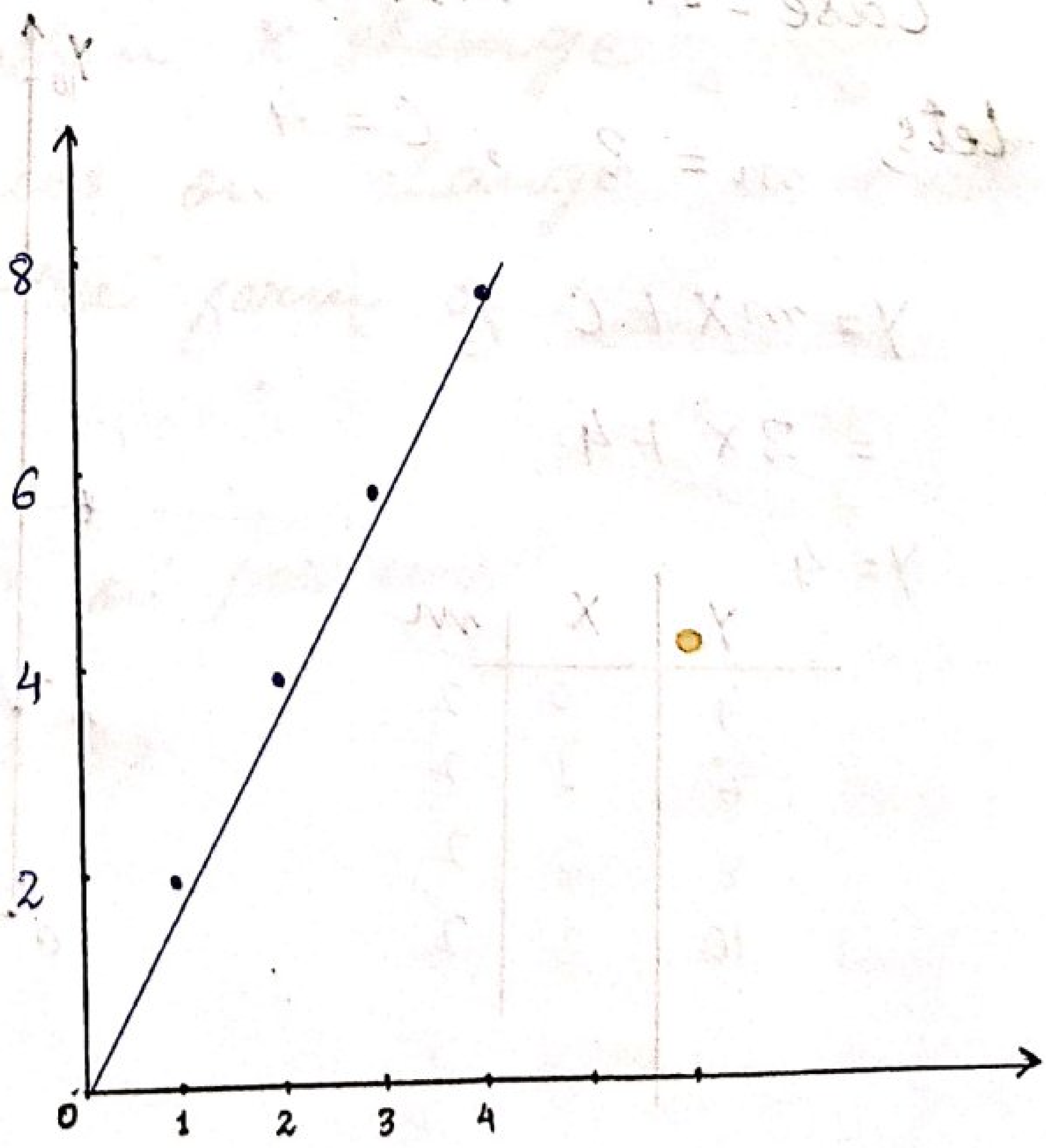
Let, $m = 2$, $C = 0$

$Y = mX + C$

$= 2X$

$Y = 0$

Y	X	m
0	0	2
2	1	2
4	2	2
6	3	2
8	4	2



— Application in Economics —

$\frac{C}{Y} = c =$ APC (Average Propensity to Consume)

$\frac{\Delta C}{\Delta Y} = c =$ MPC (Marginal Propensity to consume)

Q:- Under what condition APC = MPC.

→ When Intercept (C) zero.

3rd Case :- $m > 0$, $c < 0$

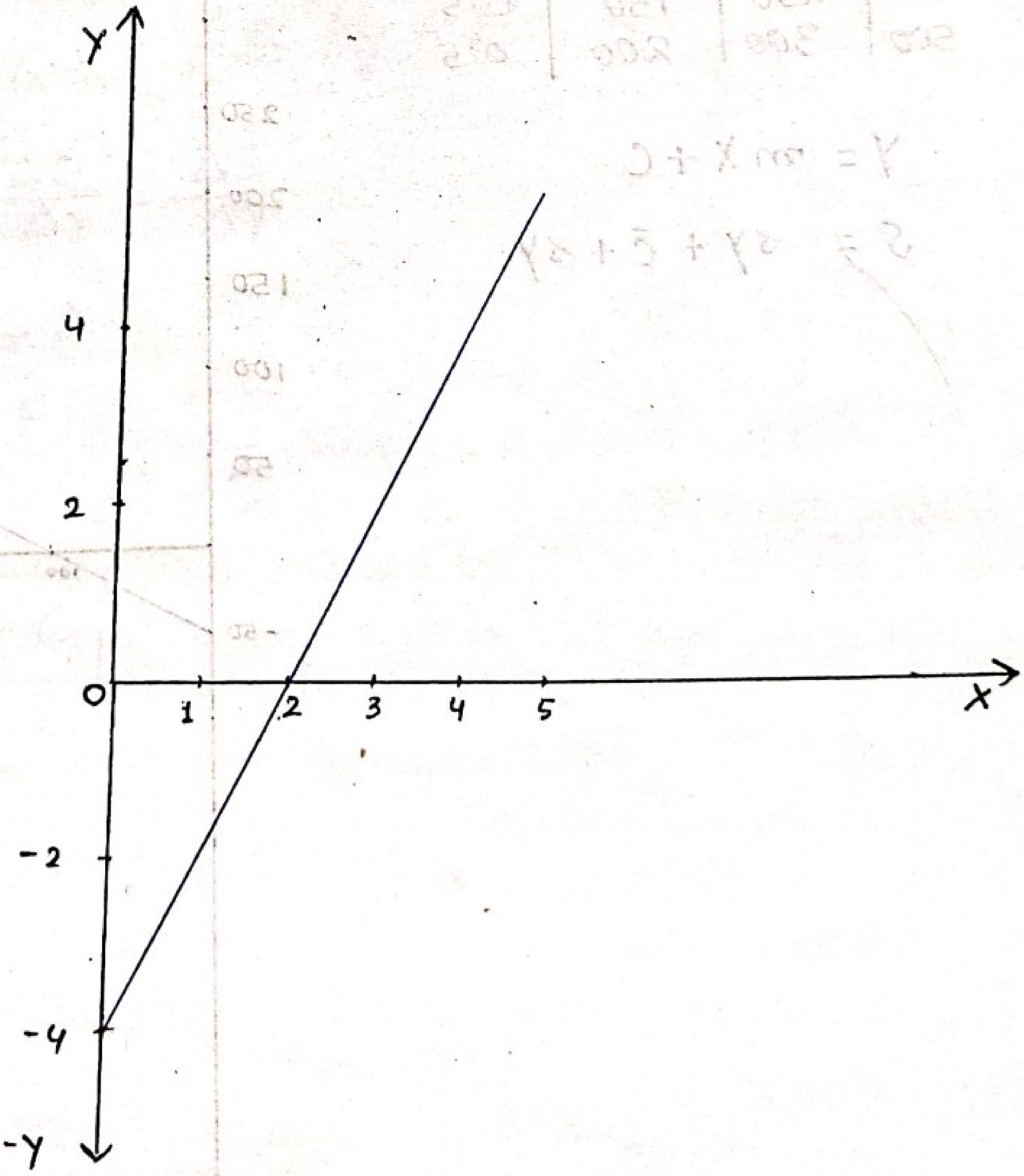
Lets, $m = 2$, $c = -4$

Y	X	m
-4	0	2
-2	1	
0	2	
2	3	
4	4	

$y = mx + c$

$= 2x \cancel{+} -4$

$y = -4$



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Application of case - 3 in Economics :—

— Saving - Income Relationship —

$S = f(Y)$

$\boxed{S = -\bar{c} + sY}$ Saving function.

Here, S = Saving

Y = Income

s = Induced Saving

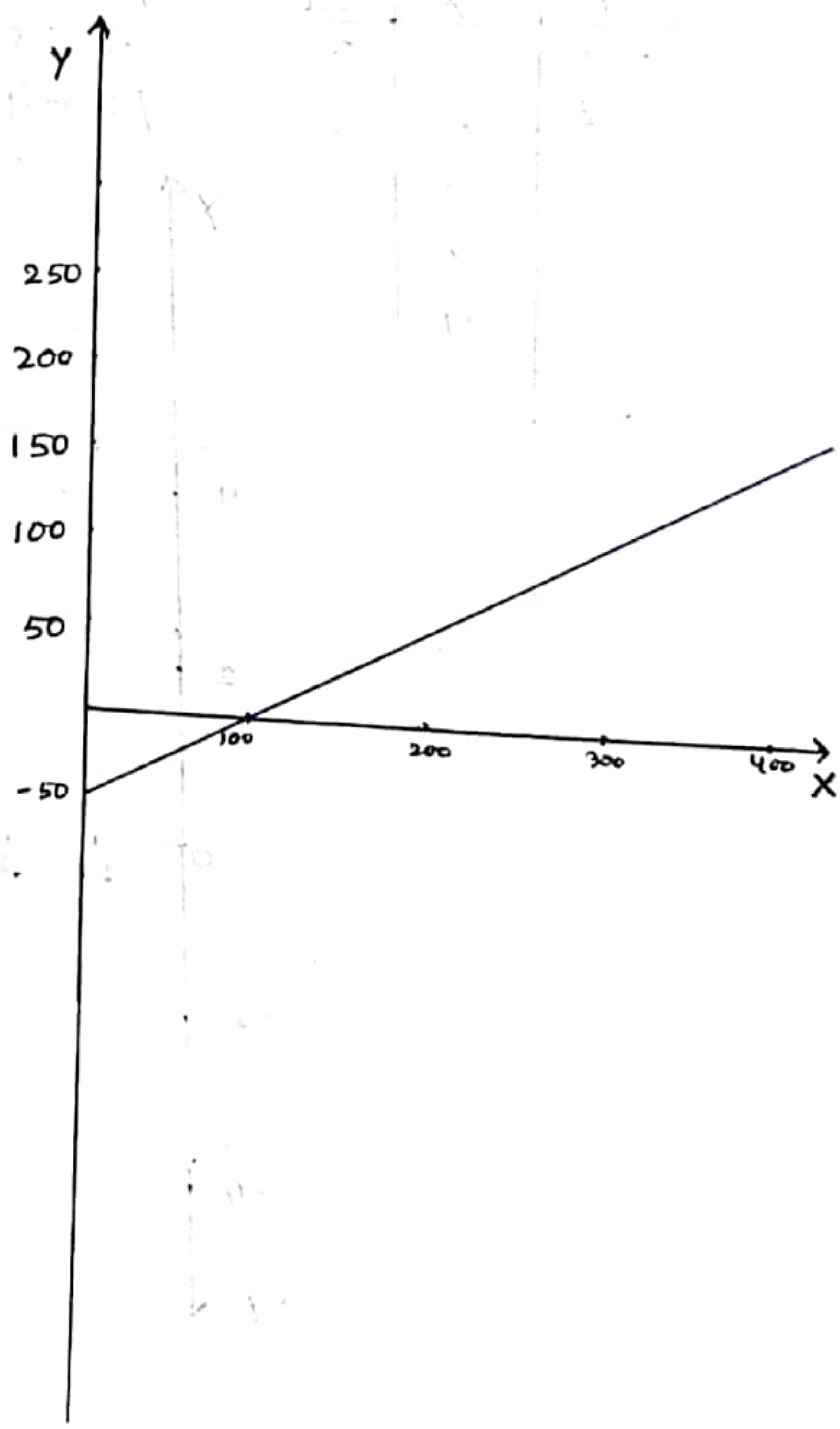
$-\bar{c}$ = Autonomous dissaving

Y	C	S	s
0	50	-50	0.5
100	100	0	0.5
200	150	50	0.5
300	200	100	0.5
400	250	150	0.5
500	300	200	0.5

$[S = Y - C]$

$Y = mX + C$

$S = sY + \bar{c} + sY$



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Derivation of saving function

$Y = C + S$ → eqn (i)

$C = \bar{C} + cY$ → (ii)

— Substituting eqn (ii) in eqn (i) —

$Y = C + S$

$Y = (\bar{C} + cY) + S$

$S = Y - (\bar{C} + cY)$

$S = Y - \bar{C} - cY$

$S = Y - cY - \bar{C}$

$S = Y(1 - c) - \bar{C}$

$S = -\bar{C} + sY$

→ Autonomous dis saving

$Y = C + S$

$\Delta Y = \Delta C + \Delta S$

Dividing both side by ΔY

$$\frac{\Delta Y}{\Delta Y} = \frac{\Delta C}{\Delta Y} = \frac{\Delta S}{\Delta Y}$$

$1 = c + s$

$s = 1 - c$

$\therefore \quad S = Y(1 - c) - \bar{C}$

$$\boxed{S = sY - \bar{C}}$$

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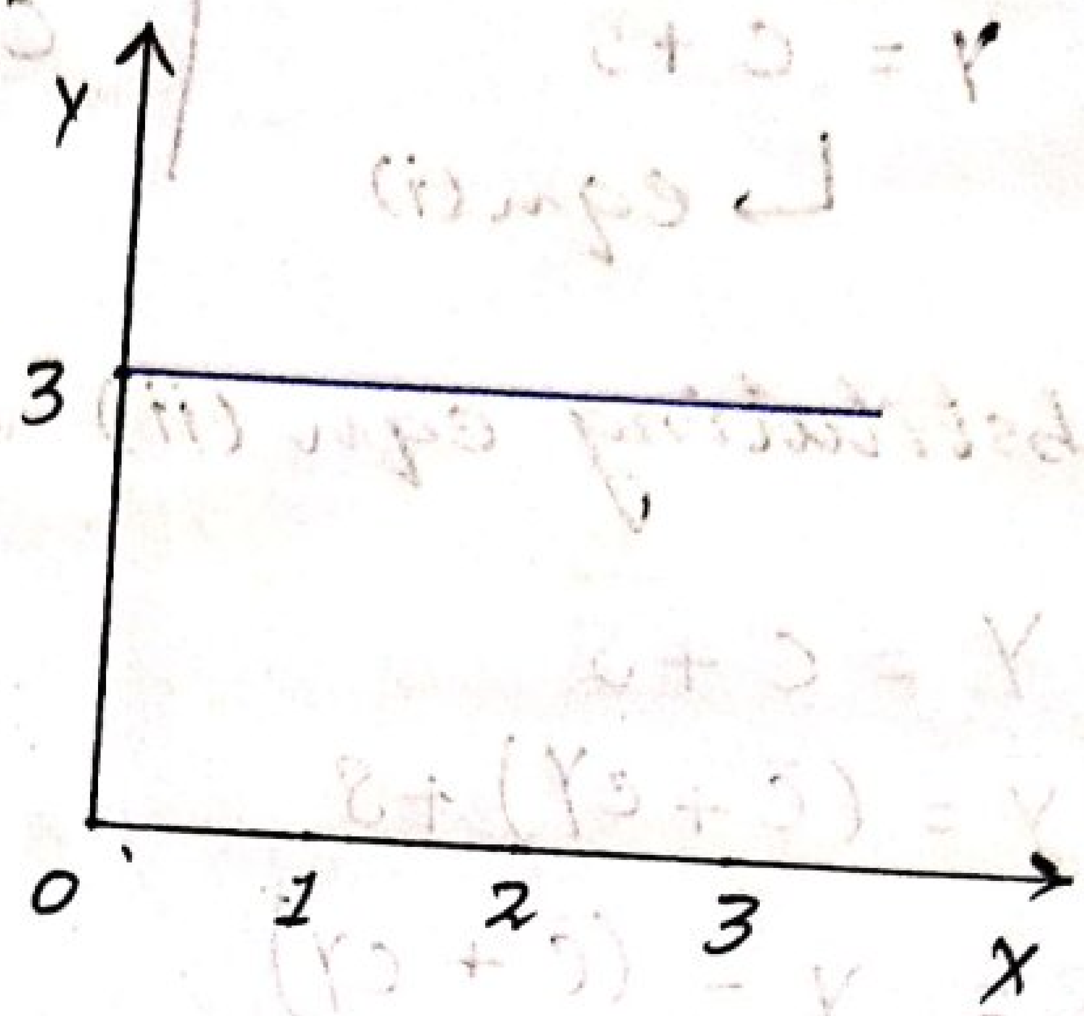
Case - 4 :- 4 $m = 0$, $c > 0$

Lets, $m = 0$, $c = 3$

$Y = mX + C$

$Y = +3$

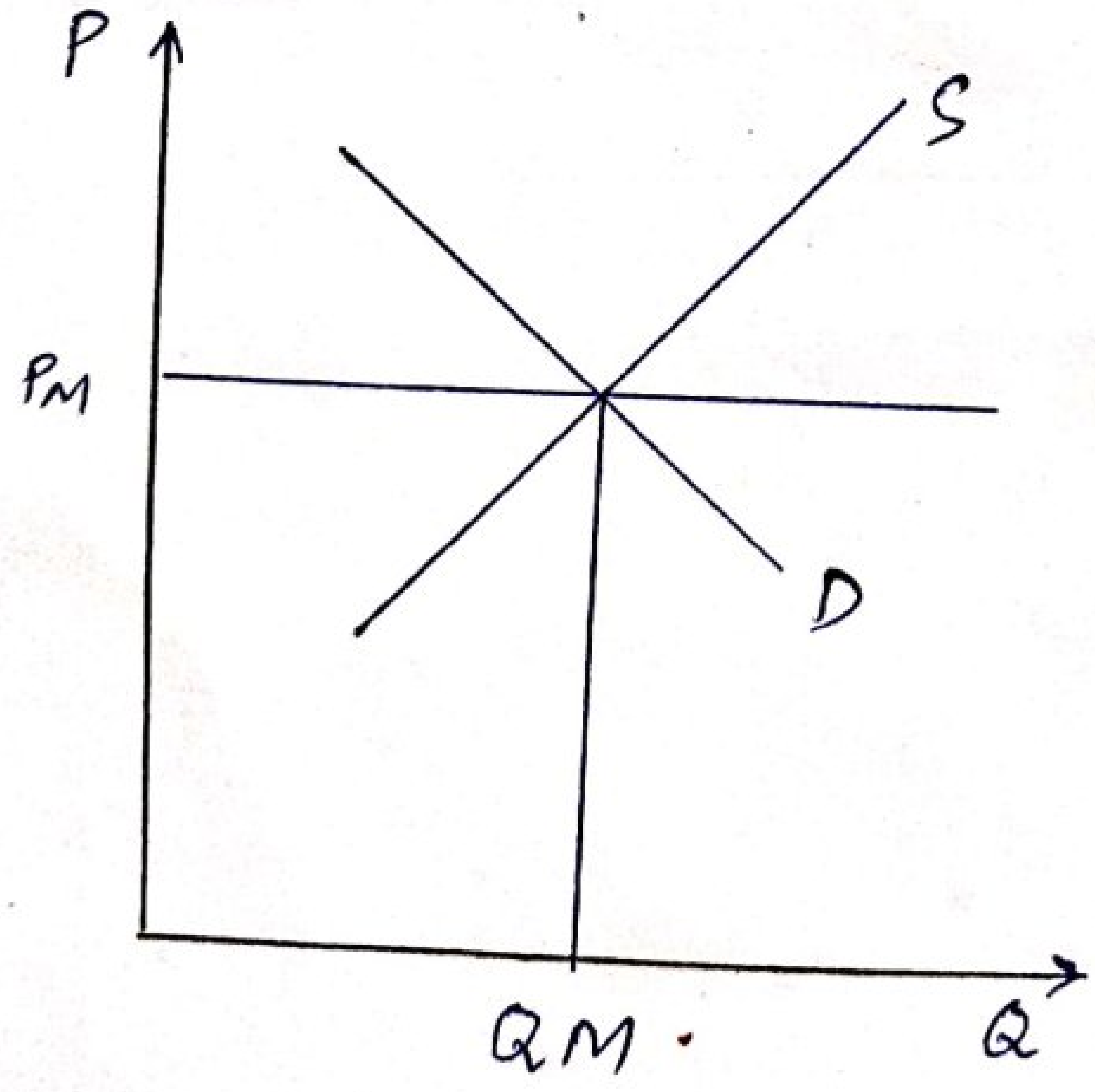
X	Y
0	3
1	3
2	3
3	3

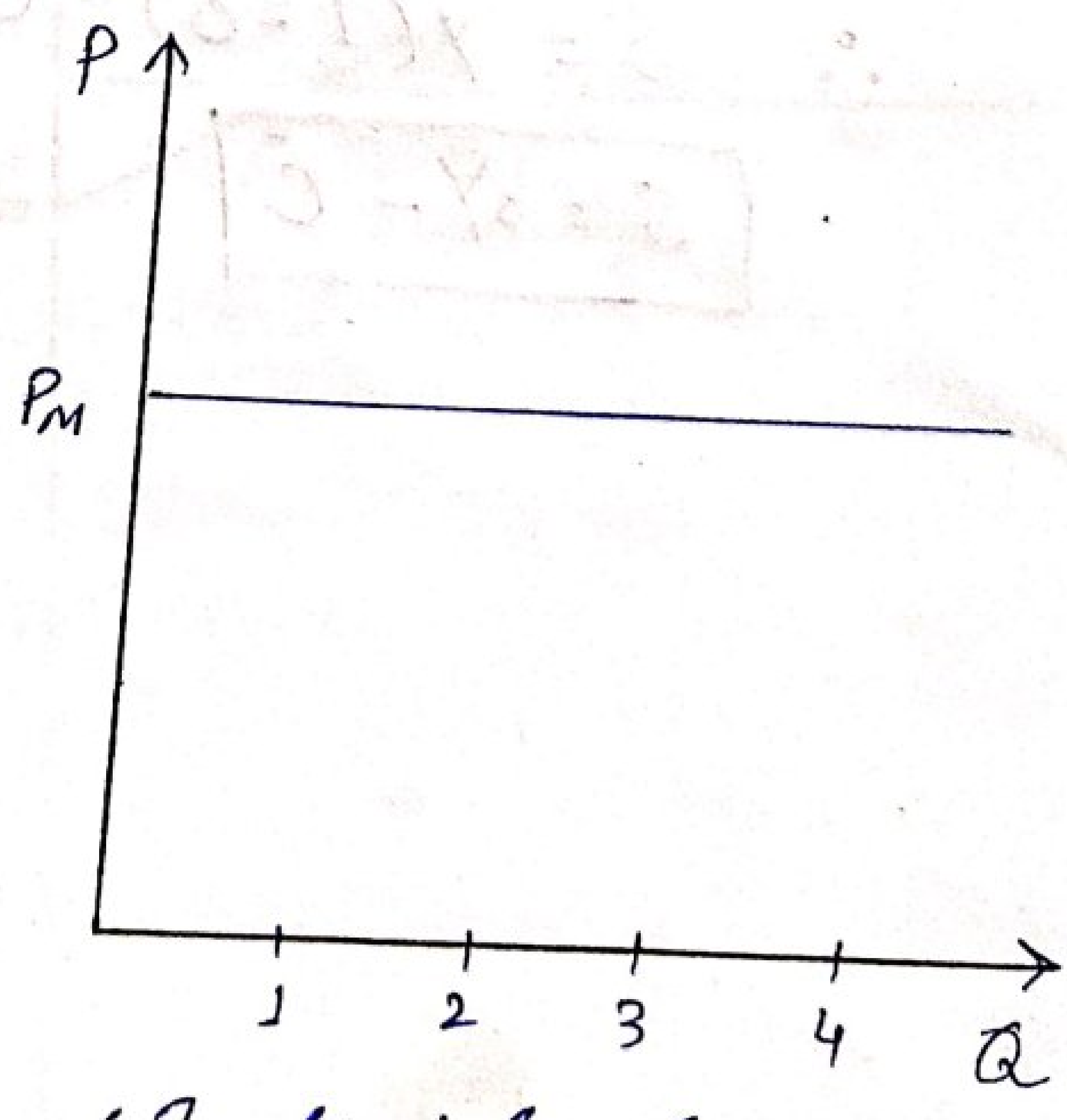


Application in Economics —

(Perfect Competetion) → charectaristics

- Large number of buyers and Sellers
- Individual firm is a price taker.
- Selling Identical / Homogenous goods.





(Individual firm)

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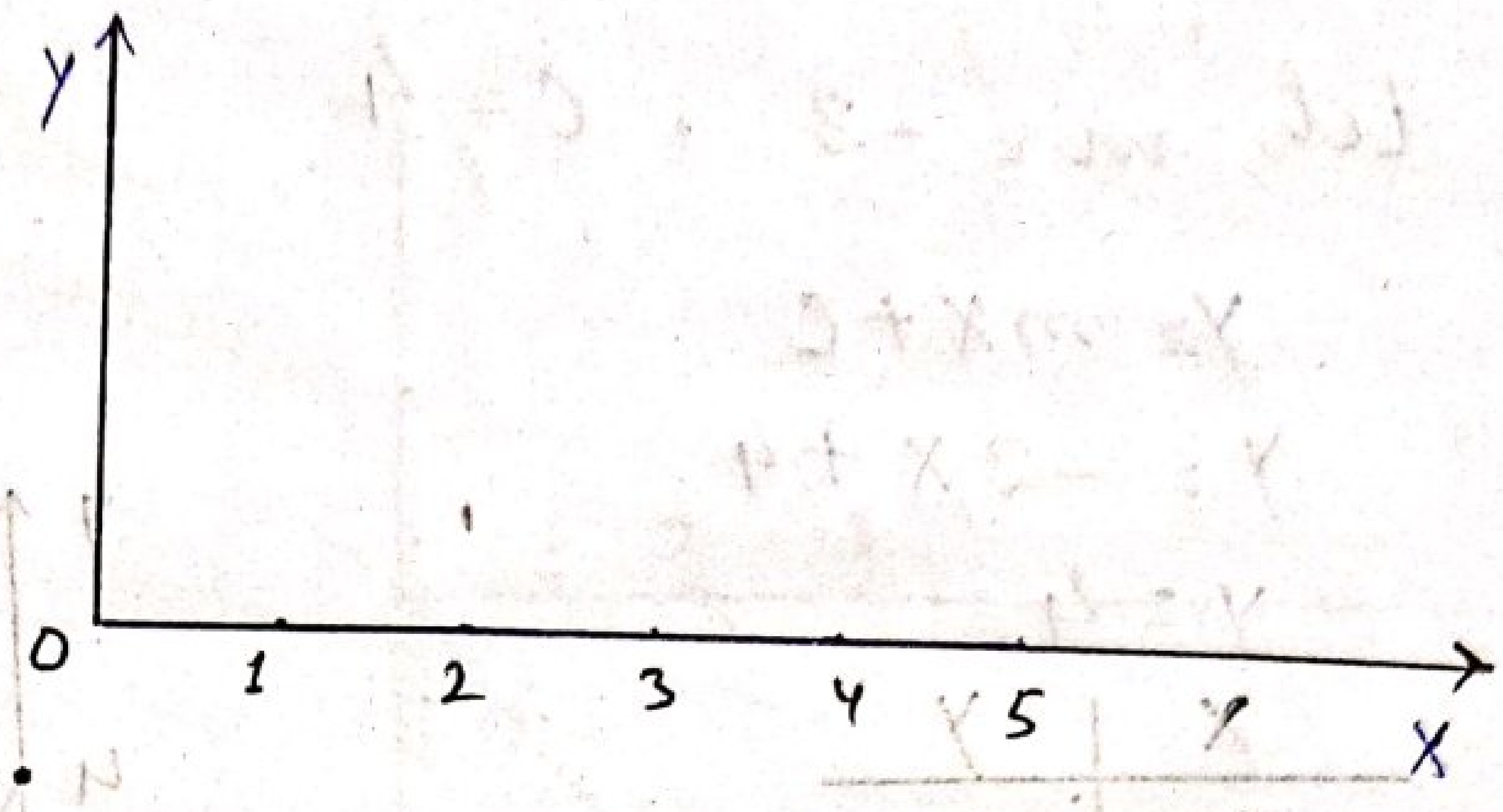
Case-5 :- $m=0$, $C=0$

X	Y
0	0
1	0
2	0
3	0
4	0
5	0



Application in Economics —

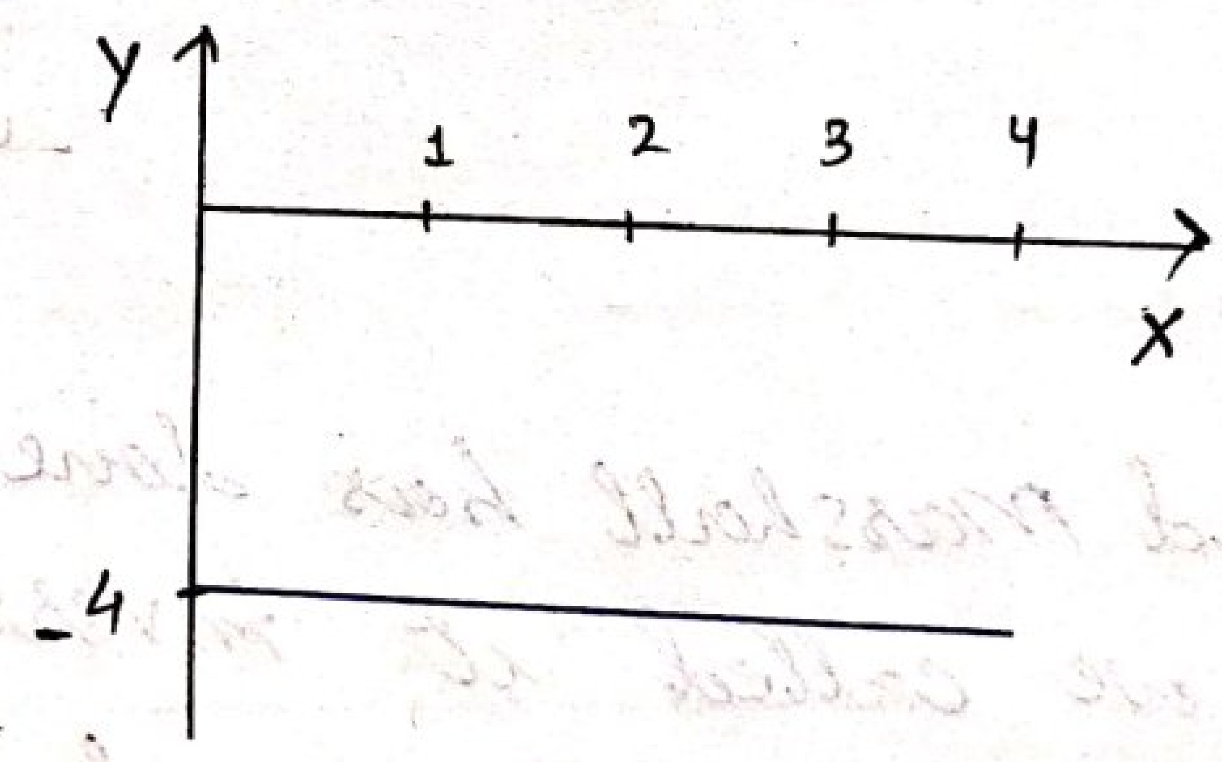
Convert Model $\frac{\Delta C}{\Delta Q} = MC = 0$

Case-6 :- $m=0$, $C<0$

Lets, $m=0$, $C = -4$

$Y = mX + C$

$= -4$



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Case - 7

$m<0, c>0$

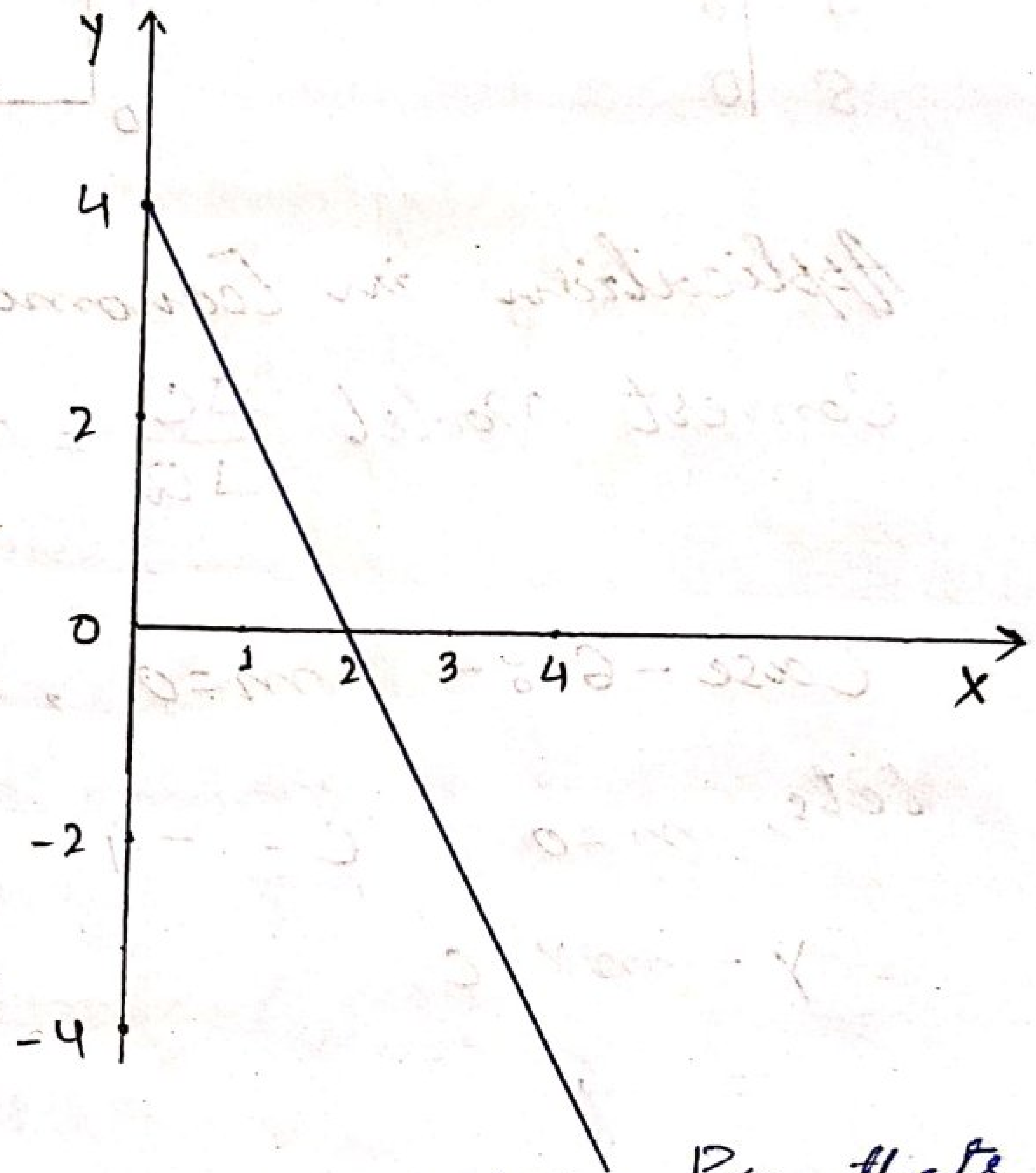
Let, $m = -2$, $c = 4$

$y = mx + c$

$y = -2x + 4$

$y = 4$

x	y
0	4
1	2
2	0
3	-2
4	-4



Alfred Marshall has done the mistake P/Q thats why we called it inverse demand function.

Demand

① feeling or desire of a person ② backed by adequate Purchasing Power.

Case - 8

$m < 0$, $c = 0$

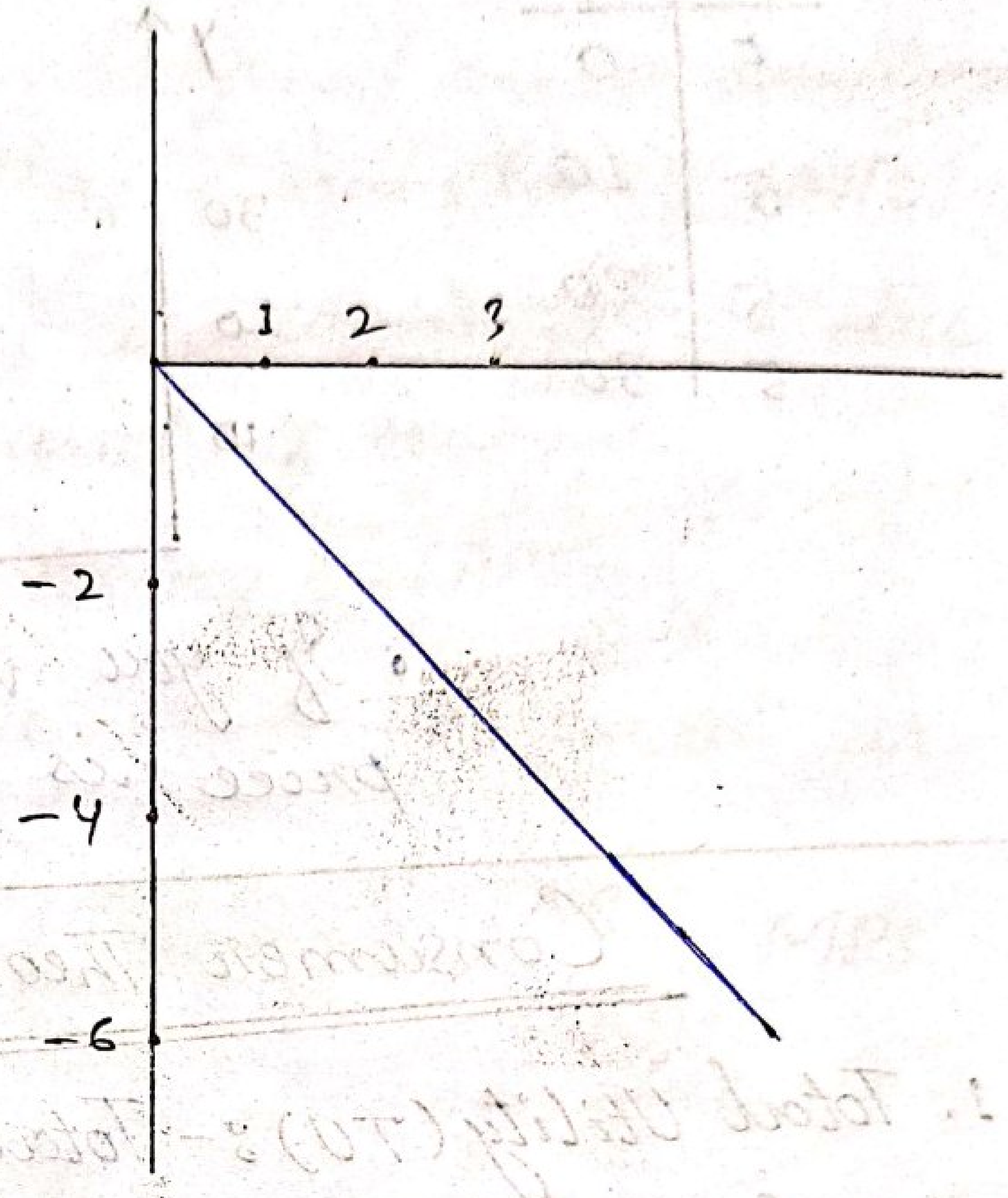
let, $m = -2$

$\therefore y = mx + C$

$y = -2x + c$

$y = 0$

x	y
0	0
1	-2
2	-4
3	-6



Case - 9

$m < 0$, $c < 0$

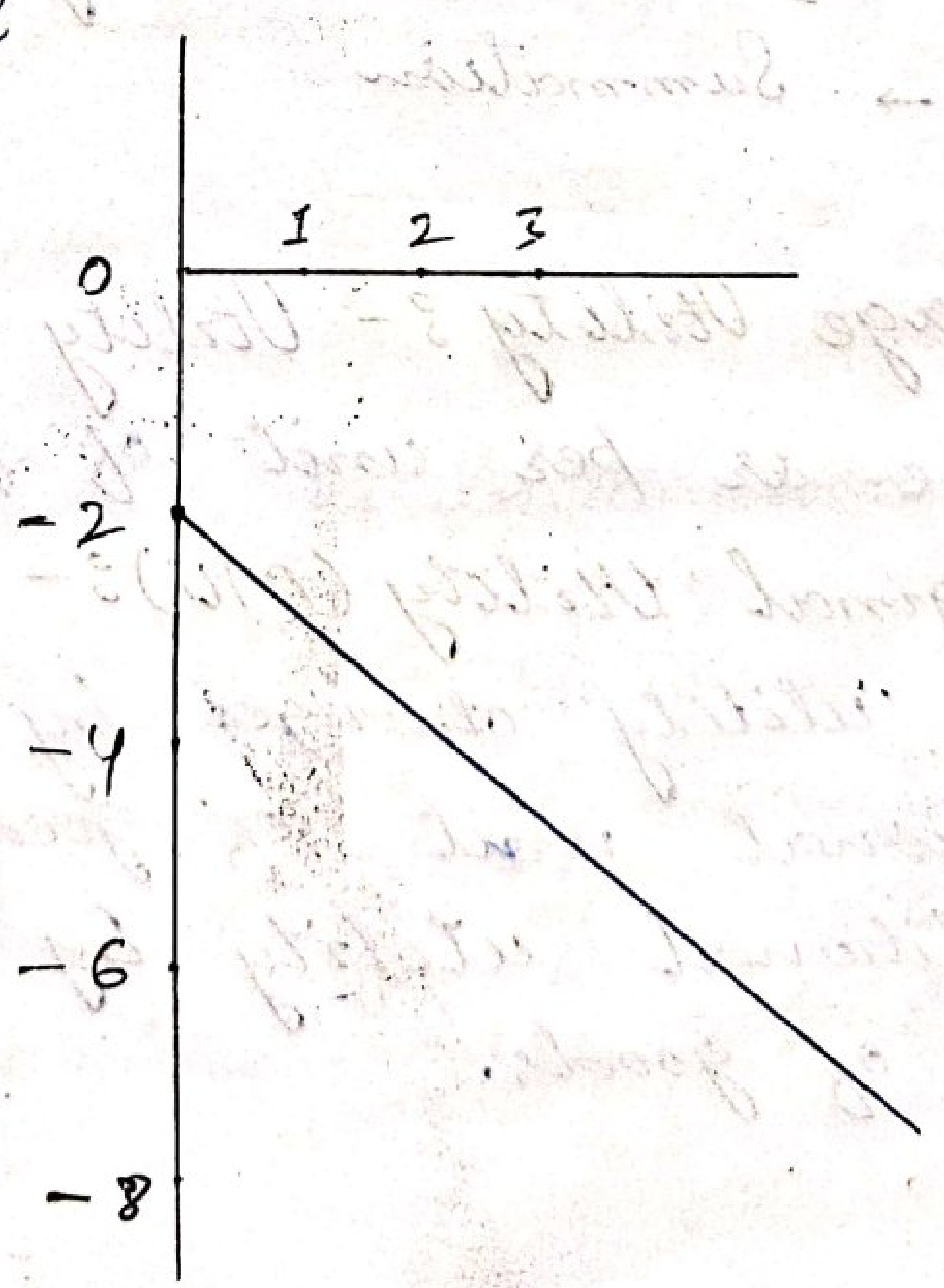
let, $m = -2$, $c = -2$

$y = mx + c$

$y = -2x - 2$

$y = -2$

x	y
0	-2
1	-4
2	-6
3	-8

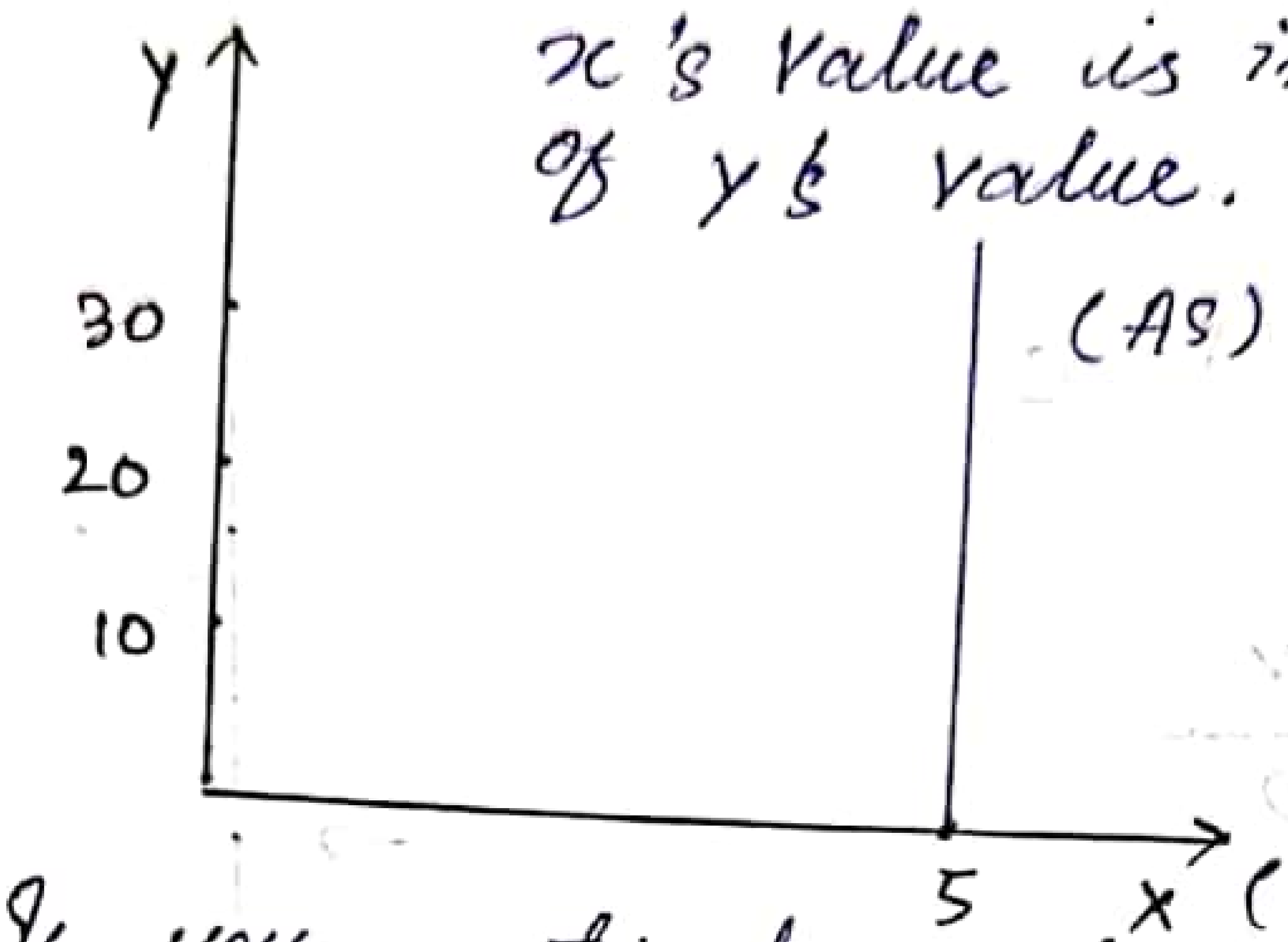


Case - 10

(Classical Model)

$m \to \infty$

X	Y
5	0
5	10
5	20
5	30



x's value is independent of y's value.

- If you vertical supply curve then price is independent of Yp.

Consumer Theology

1. Total Utility (TU) :- Total satisfaction obtained or derived by consuming goods (at that perticular time period)

→ In other words, it means some of utilities derived from consuming a good.

$\sum_{i=1}^{n} u_i \to$ Summation [used in... Reveal Preference Theory]

2. Average Utility :- Utility obtained by consuming average or per unit of goods.

3. Marginal Utility (MU) :- $\frac{\Delta TU}{\Delta Q}$ How much total utility changes by consuming 1 additional unit of goods.

→ Additional utility by consuming 1 more unit of goods.

Consumption — Income Relationship

Propensity =

Average Propensity of Consumption :- $(APC) = \frac{C}{Y}$

→ Proportion of Income Spend on consumption.

Marginal Propensity of consumption :- $(MPC) = \frac{\Delta C}{\Delta Y}$

→ How much changes in consumption due to an additional unit of income.

Average Propensity of Saving :- $(APS) = \frac{S}{Y}$

→ Proportion of income that is saved or consumed is called APS.

Marginal Propensity of Saving :- $\frac{\Delta S}{\Delta Y} = MPS$

→ How much change in saving due to an additional unit of Income.

Producer Theology

1. Total Product (TP) :- Total quantity of output produced.
2. Average Product (AP) :- $\frac{TP}{L}$ = Output obtained per unit of labour.

↳ In Micro economics   ↳ In macroeconomics

$\frac{Q}{L}$ (Productivity of labour) $\frac{Y}{L}$

→ Increase in productivity of labour, it means, same productivity of labour.

→ Each labour produce more output.

3. Average Product (AP_K) → with perspective of capital

$\frac{TP}{K}$ = Output produce by per unit of capital

4. Marginal Product (MP_L) → with respect of labour

$\frac{\Delta TP}{\Delta L}$ = How much change in total product by increasing labour.

↳ Additional output obtained by employing one additional unit of labour.

(Here K is constant)

In Microeconomics $\Delta Q / \Delta L$

In Macro... $\Delta Y / \Delta L$

5) Marginal Product (MP_K) → with respect of capital.

→ In this labour remain constant

↳ How much total product changes by an additional unit of capital → $\frac{\Delta TP}{\Delta K}$

Total cost (TC):- Total outlay of resources to produce output.

$C = wage(w) \times l + VK$ (Vental rate of capital)

Variable cost

Fixed cost

V = output

wl (labour cost)

VK (capital cost)

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→ $TC = TFC + TVC$

→ Total cost is a sum of fixed cost & variable cost.

Average cost (AC) :- $\frac{TC}{Q}$ = It is per unit cost of production.

Marginal Cost (MC) :- $\frac{\Delta TC}{\Delta Q}$ = How much change in total cost by 1 additional unit of quantity.

$$MC = \frac{\Delta TC}{\Delta Q} = \frac{\Delta TFC}{\Delta Q} + \frac{\Delta TVC}{\Delta Q}$$

→ How much changes in total variable cost by adding an additional unit of quantity with the respect of no change in FC.

$AC \rightarrow AFC + AVC$:- It is the sum of "Average" fixed cost and 'Average' variable cost.

$$AC = AFC + AVC$$

It varie with output.

Fixed cost per unit of Quantity

It varie with output

Per unit cost of output.

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Revenue Concept

Total Revenue (TR) = Price . Quantity (P.Q)

→ Total income / revenue earn from sale of goods.

Average Revenue (AR) = (Price) → $\frac{TR}{Q} = \frac{P.Q}{Q} = P$

Marginal Revenue (MR) = $\frac{\Delta TR}{\Delta Q}$ = How much total revenue changes when output sold an additional unit.

Note :- In consumer perspective we called it expenditure.

Profit

π (Profit) = TR − TC [Total Revenue − Total cost]

$A\pi = \frac{\pi}{Q}$

$MP(\pi) = \frac{\Delta \pi}{\Delta Q}$

Note :- Two parellel line must have equal slope.

→

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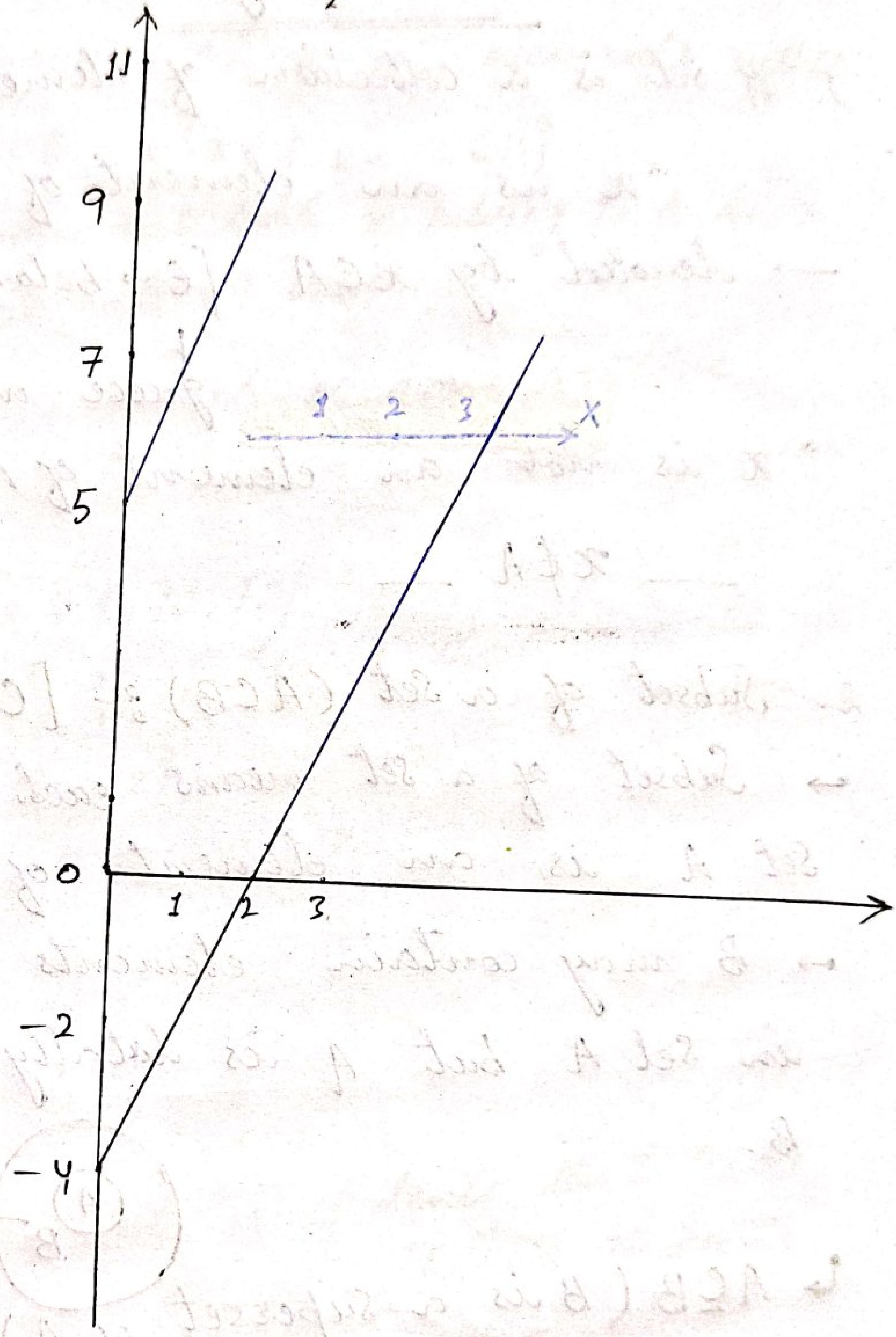
Example – $Y_1 = 2x - 4$, $Y_2 = 2x + 5$

X	Y_1	Y_2
0	-4	5
1	-2	7
2	0	9
3	2	11



Set Theory

1. A set is a collection of elements.

"x is an element of A"

→ denoted by $x \in A$ [$\in$ → belong to]

↓ greece word epsilon.

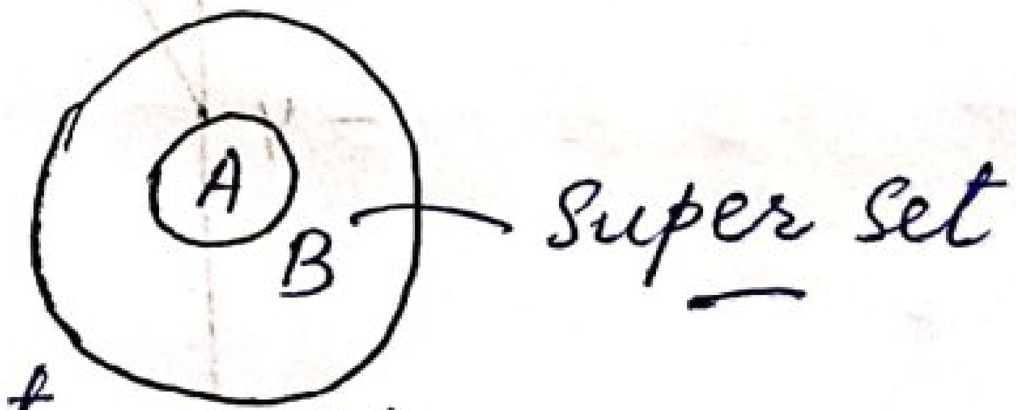
"x is not an element of A"

— $x \notin A$ —

2. Subset of a Set ($A \subset B$) :- [$\subset$ → Proper Superset]

→ Subset of a set means each elements of Set A is an element of set B

→ B may contain elements which are not in Set A but A is totally contain within B.



↳ $A \subseteq B$ (B is a superset of A)

↳ $A \subset B$ (B is a proper super set of A)

Example :-

A = Set of odd positive integers.

B = Set of Positive integers.

Here, $A = 1, 3, 5, 7$; $B = 1, 2, 3, 4, 5, 6, 7, 8$

So, B is a proper superset of A $A \subset B$